

Certified coexistence of symmetry-reduced local maximizers in a periodic 3D Navier–Stokes Fourier class

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Abstract

We study the classical fixed- enstrophy variational problem for instantaneous enstrophy production in periodic three-dimensional Navier–Stokes flow, within a prescribed 16-pair Fourier class with 64 real divergence-free coordinates. We give two exact symmetry reductions. The first is a six-amplitude fixed space on which a Sturm and interval-arithmetic calculation proves the global maximum of the static rate. Its high-parameter branch is global within that symmetry class above an explicit threshold. The second is a two-amplitude fixed space with a closed-form stationary branch and a closed optimized-rate formula. Symbolic algebra verifies that it is a full 64-coordinate KKT branch for every positive enstrophy and viscosity. A rational interval cover with separate 256- and 512-bit Arb calculations certifies this branch as a strict full-space local maximum modulo translations for every such parameter pair. The high-parameter six-amplitude branch is independently certified as a strict full-space local maximum throughout its entire existence range. At the inviscid endpoint, the explicit six-amplitude limiting field is also certified as a strict full-space local maximum of stretching, modulo translations; together with the competing branch it gives two distinct exact full-space local maxima with an explicit value gap. An exact rational sum-of-squares certificate also controls the $(1, 1, 2)$ shell component of the stretching cubic, while leaving cross-component alignment open. At 10,000 it exceeds the six-amplitude rate. The two symmetry branches cross exactly once at an interval-certified dimensionless parameter value near 27,993. All statements are finite dimensional; none addresses global regularity or singularity formation for the Navier–Stokes PDE.

Key words. Navier–Stokes equations; enstrophy production; computer-assisted proof; interval arithmetic; Fourier–Galerkin methods.

MSC codes. 76D05, 65G40.

1 Introduction

For a periodic incompressible velocity field, the instantaneous production of enstrophy is a natural diagnostic for extreme three-dimensional flows. The fixed- enstrophy variational problem was studied numerically by Lu and Doering [6] and by Ayala and Protas [1]; those works established important extreme-flow branches but did not give an exact finite- enstrophy , computer-assisted classification of a prescribed Fourier class. We give such a classification for two symmetry-fixed subspaces of a 64-dimensional Fourier polynomial.

The novelty claimed here is deliberately narrow: exact Fourier reductions, exact algebraic branch formulas, and computer-assisted local maximality in a specified finite class. This differs from numerical optimization of the unrestricted PDE problem [6, 1, 2] and from Fourier-restricted model equations with a modified projection [7], as well as recent orbit-level Galerkin analyses [3, 4]. The finite-dimensional result should not be read as a statement about Navier–Stokes regularity.

2 Static Fourier variational problem

On $\mathbb{T}^3 = [0, 2\pi)^3$, let

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad \nabla \cdot u = 0.$$

With $\omega = \nabla \times u$ and $S = (\nabla u + \nabla u^T)/2$, set

$$\mathcal{E} = \frac{1}{2} \langle |\omega|^2 \rangle, \quad P = \frac{1}{2} \langle |\nabla \omega|^2 \rangle, \quad A = \langle \omega \cdot S \omega \rangle, \quad \mathcal{R} = A - 2\nu P.$$

We restrict u to the canonical divergence-free polarization basis on the 16 independent pairs $0 < |k|^2 \leq 4$, yielding $x \in \mathbb{R}^{64}$. Our static problem is $\max\{\mathcal{R}(x) : \mathcal{E}(x) = E_0\}$.

2.1 Exact Fourier and symmetry conventions

For reproducibility, we state the finite-dimensional model precisely. Let $\mathcal{K}$ contain the lexicographically positive representative of every pair $\{k, -k\}$ with $0 < |k|^2 \leq 4$. Thus $|\mathcal{K}| = 16$. For each $k \in \mathcal{K}$, choose the coordinate vector e_j whose corresponding entry of k has least absolute value (breaking ties by j), and set

$$p_1(k) = k \times e_j, \quad p_2(k) = k \times p_1(k).$$

The real and imaginary coefficients of $p_1(k)$ and $p_2(k)$ give the four real coordinates for the pair, while $\widehat{u}(-k) = \overline{\widehat{u}(k)}$. These integer polarizations fix the 64-coordinate convention used in every theorem and archive.

All convolution, Leray projection, and inner products in the certificates are performed over $\mathbb{Q}(i)$. In particular, if $B(v, w) = -\mathbb{P}[(v \cdot \nabla)w]$, then the cubic term is evaluated as $A = \langle \nabla \times u, \nabla \times B(u, u) \rangle$ by exact finite Fourier sums. There is no floating-point identification of the displayed reduced polynomials.

For a signed-coordinate orthogonal matrix Q and $m \in \mathbb{Z}^3$, write

$$(g_{Q,m}u)(x) = Qu\left(Q^T\left(x - \frac{\pi m}{2}\right)\right).$$

On Fourier coefficients this is

$$\widehat{g_{Q,m}u}(k) = Q\widehat{u}(Q^T k) \exp\left(-i Q^T k \cdot \frac{\pi m}{2}\right).$$

This action preserves $\mathcal{E}$, P , and A . We form common fixed spaces by exact rational row reduction of $g_{Q,m} - I$ in the above 64 coordinates.

2.2 An exact shellwise SOS bound

Write E_j for the enstrophy carried by the $|k|^2 = j$ shell, and let $A_{112}, A_{123}, A_{224}, A_{334}$ denote the four parts of the exact stretching cubic, indexed by their squared-shell patterns. The following finite-class estimate is a useful check on the exact cubic algebra.

Proposition 2.1 (A rational SOS shell bound). *In the canonical 16-pair Fourier class,*

$$\begin{aligned} A_{112}^2 &\leq \frac{4}{3} E_1^2 E_2, & |A_{112}| &\leq \frac{2}{\sqrt{3}} E_1 \sqrt{E_2}, \\ A_{123}^2 &\leq 16 E_1 E_2 E_3, & A_{224}^2 &\leq E_2^2 E_4, \\ |A_{224}| &\leq E_2 \sqrt{E_4}, & A_{334}^2 &\leq \frac{2}{27} E_3^2 E_4. \end{aligned}$$

Exact certificate. For each pattern, write the cubic as $\sum_j M_j(y)z_j$ in the coordinates of the repeated and linear shells. Weighted Cauchy reduces the claim to a quartic in y . The $(1, 1, 2)$ quartic is represented on 78 degree-two monomials by a rational Gram matrix with 40 positive Schur pivots and a 38-dimensional zero remainder. The $(2, 2, 4)$ quartic has a separate rational Gram matrix on 300 degree-two monomials, with 106 positive pivots and a 194-dimensional zero remainder. In both cases exact coefficient expansion agrees with the Fourier cubic, hence $G \succeq 0$ over $\mathbb{Q}$. The other two patterns are checked by separate exact Gram identities: a 288-by-288 positive-definite bilinear Gram matrix for $(1, 2, 3)$, and a sparse rank-three 136-by-136 Gram matrix for $(3, 3, 4)$. These are bounds for individual triad components only: they neither prove sharp constants nor bound the full stretching cubic. $\square$

2.3 Certified local statements

For a phase slice, the constrained stationarity equations consist of the 64 rate equations, the enstrophy constraint, and three translation phase conditions. The resulting 68-dimensional polynomial system is evaluated with outward-rounded Arb balls. Strict inclusion of its Krawczyk image in a box proves a unique zero in that box [5]. An exact-dyadic congruence followed by interval Gershgorin bounds gives the bordered inertia. The reported inertia $(4, 64, 0)$ proves nondegeneracy and the strict constrained local maximum modulo the three translations. This is a local statement in the stated Fourier class only.

3 Six-amplitude branch

Three combined signed-coordinate and quarter-period translation symmetries have a six-dimensional common fixed space. Exact rational row reduction identifies coordinates (a, b, c, d, e, f) in which

$$A = 64acd + 32aef - 64bdf, \quad (1)$$

$$\mathcal{E} = 2a^2 + 32b^2 + 48c^2 + 8d^2 + 48e^2 + 4f^2, \quad (2)$$

$$P = 2a^2 + 128b^2 + 144c^2 + 16d^2 + 144e^2 + 8f^2. \quad (3)$$

Theorem 3.1 (Certified ansatz globality). *At $\nu = 10^{-2}$ and $\mathcal{E} = 100$, the constrained maximum of $\mathcal{R}$ in the six-amplitude fixed space is unique after the finite sign gauge and equals*

$$449.718337230445715677500657 \dots$$

The corresponding full 64-coordinate point is a strict local maximum modulo translations.

The proof combines exact face exhaustion, Sturm isolation of the interior multiplier root, and separate 256/512-bit Arb comparisons. Symmetric criticality promotes restricted stationarity to full-space stationarity [8]; a 68-variable Krawczyk inclusion and certified bordered inertia $(4, 64, 0)$ give the local statement. The interval inclusion is used in its usual existence-and-uniqueness role [5].

Writing $\eta = \mathcal{E}/\nu^2$ and $\mu = \lambda/\nu$, the high- η interior branch satisfies

$$69\mu^4 + 1696\mu^3 + (14936 - 32\eta)\mu^2 + (54656 - 512\eta)\mu + (68368 - 2048\eta) = 0. \quad (4)$$

Descartes' rule and an exact sign bound at $\sqrt{28}$ imply a unique admissible positive root for $\eta > 1867/4$. Positive-coefficient face resultants prevent crossing of the interior and face branches, hence the interior branch is global inside this symmetry class throughout that range.

Theorem 3.2 (High-parameter six-amplitude branch). *For every $\eta > 1867/4$, the quartic (4) has one admissible positive interior root. The associated stationary point is the global maximizer of $\mathcal{R}$ within the six-amplitude fixed space, after the finite sign gauge. This theorem makes no full 64-coordinate global assertion.*

Theorem 3.3 (Uniform full-space stability of the high- η six branch). *For every $\eta > 1867/4$, the branch in Theorem 3.2 is a strict local maximizer of $\mathcal{R}$ subject to fixed $\mathcal{E}$, modulo translations, in the complete 64-real-coordinate Fourier class. This is not a full-space global-maximality statement.*

Certificate outline. Put $r = \mu^{-1} = s/10$. At $\eta = 1867/4$, the left side of (4) at $\mu = 10$ equals -344736 ; uniqueness of the positive root therefore gives $0 < s < 1$ throughout the physical branch. On the compact auxiliary cover $0 \leq s \leq 1$, exact radical branch formulas define the full bordered Hessian after a fixed quarter-period translation to a regular three-phase slice. At the midpoint of each of 2,048 rational boxes, Arb certifies bordered inertia $(4, 64, 0)$. The Frobenius-norm bound $\|B(s_0)^{-1}(B(s) - B(s_0))\|_F < 1$ on each box gives a nonsingular symmetric homotopy, hence preserves inertia across the box. Independent 256- and 512-bit covers close. The source-bound record is `six_branch_high_eta_uniform.json`. $\square$

4 A competing two-amplitude branch

Two different lattice symmetries have a two-dimensional common fixed space. They are

$$(Q_1, m_1) = (\text{diag}(-1, 1, 1), (0, 2, 0)), \quad (Q_2, m_2) = \left(\begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & 0 \end{pmatrix}, (0, 0, 0) \right).$$

Exact row reduction gives fixed-space dimension two; the two displayed generators themselves, rather than a numerically discovered sparsity pattern, define the ansatz. Their exact finite closure has 24 elements and is internally isomorphic to $A_4 \times C_2$: it has a central order-two inversion-translation and an A_4 complement. This group identification is obtained by exact multiplication of signed-coordinate matrices and quarter-period translations. It is a structural description of the ansatz symmetry, not a parameter-uniform stability claim. Its exact action on the 64 real controls has isotypic ranks $(2, 6, 24)$ in each of the two central- C_2 parity sectors; this decomposition is included to expose the block structure relevant to local stability, but no uniform Hessian-sign conclusion is drawn here. In the canonical basis, three coordinates equal a , six equal signed copies of b , and all other coordinates vanish. Exact convolution gives

$$A = 24a^2b, \quad \mathcal{E} = 3a^2 + 48b^2, \quad P = 3a^2 + 96b^2. \quad (5)$$

Recovering the entire restricted quadratic/cubic coefficient table directly from the full rational 64-mode convolution gives respectively $(3, 0, 48)$, $(3, 0, 96)$, and $(0, 24, 0, 0)$ in the monomial orders (a^2, ab, b^2) and (a^3, a^2b, ab^2, b^3) . Thus (5) is an identity of polynomials, not an interpolation through candidate points. The same exact Fourier calculation gives kinetic energy and helicity

$$K = 3a^2 + 24b^2, \quad H = \langle u \cdot \nabla \times u \rangle = 0.$$

Thus the entire competing symmetry plane is nonhelical; this is a structural description of this branch, not a general feature of enstrophy extremizers. Moreover, the a modes lie entirely on the $|k|^2 = 1$ shell and the b modes entirely on the $|k|^2 = 2$ shell. The exact cubic $24a^2b$ is therefore a

specific nonhelical two-shell interaction, rather than a diffuse low-mode superposition. Equivalently, the branch has the explicit real-space representative

$$u_{a,b}(x, y, z) = \begin{pmatrix} -2a \cos y - 8b \sin y \cos z \\ 2a \cos z + 8b \cos x \sin z \\ 2a \cos x - 8b \sin x \cos y \end{pmatrix}. \quad (6)$$

Thus the formula can be checked directly in physical space or from the 18 Fourier coefficients archived by the standalone rational audit. Eliminating a yields

$$\mathcal{R}(b) = 8\mathcal{E}b - 384b^3 - 2\nu\mathcal{E} - 96\nu b^2, \quad 0 \leq b \leq \sqrt{\mathcal{E}/48}.$$

Its derivative is strictly decreasing, so the unique sign-gauged maximizer is

$$b_* = \frac{\sqrt{\mathcal{E} + \nu^2} - \nu}{12}, \quad a_*^2 = \frac{\mathcal{E} - 48b_*^2}{3}. \quad (7)$$

Substitution gives a closed formula for the optimized rate on this plane,

$$\mathcal{R}_*(\mathcal{E}, \nu) = \frac{4}{9} \left[(\mathcal{E} + \nu^2)^{3/2} - 6\mathcal{E}\nu - \nu^3 \right]. \quad (8)$$

For $\mathcal{E}, \nu > 0$, it is positive precisely when

$$\frac{\mathcal{E}}{\nu^2} > \frac{33 + 15\sqrt{5}}{2}.$$

Theorem 4.1 (Exact optimization in the competing plane). *For every $\mathcal{E}, \nu > 0$, after fixing the inessential sign of a , the branch (7) is the unique global maximizer of $\mathcal{R}$ subject to $E(a, b) = \mathcal{E}$ within the two-amplitude symmetry plane. This is a statement only about that plane, not the unrestricted Fourier class.*

Proof. For a fixed positive $\mathcal{E}$, eliminate a by $a^2 = (\mathcal{E} - 48b^2)/3$. A global maximizer may be sign-gauged to $b \geq 0$, where $0 \leq b \leq \sqrt{\mathcal{E}/48}$ and

$$\mathcal{R}(b) = 8\mathcal{E}b - 384b^3 - 2\nu\mathcal{E} - 96\nu b^2.$$

Its derivative $8\mathcal{E} - 1152b^2 - 192\nu b$ is strictly decreasing, positive at zero, and negative at the other endpoint. Thus it has precisely one interior zero, which is (7); this is the global maximum on the interval. For $b \geq 0$, the direct comparison

$$\mathcal{R}(b) - \mathcal{R}(-b) = 16b(\mathcal{E} - 48b^2) \geq 0$$

handles the opposite half of the constraint circle. □

Direct symbolic reduction of every one of the 64 KKT equations shows that (7), with multiplier $\lambda = 8b - 2\nu$, is a full-coordinate KKT branch for all $\mathcal{E}, \nu > 0$. Scaling controls by $\sqrt{\mathcal{E}}$ reduces the local constrained problem to unit enstrophy with parameter $\eta = \mathcal{E}/\nu^2$. On the exact branch, write $s = 12b$. Then

$$a = \frac{\sqrt{3 - s^2}}{3}, \quad \frac{\nu}{\sqrt{\mathcal{E}}} = \frac{1 - s^2}{2s}, \quad \eta = \frac{4s^2}{(1 - s^2)^2}.$$

This last expression maps $0 < s < 1$ monotonically onto all positive η . There is also an exact spectral reduction behind the local calculation. Let D_E be the diagonal enstrophy quadratic form

and let H_s be the Hessian of $\mathcal{R} - \lambda\mathcal{E}$ at this branch, with $\lambda = 5s/3 - 1/s$. Since the lattice symmetry group is $A_4 \times C_2$, the generalized operator $D_E^{-1}H_s$ preserves its six real isotypic sectors. Exact symbolic Fourier algebra gives, up to a nonzero scalar,

$$\chi_s(t) = p_{0,+}p_{2,+}^2p_{3,+}^3p_{0,-}p_{2,-}^2t^3(7s^2 - 3st - 9)^3p_{3,-}^3. \quad (9)$$

Here $p_{0,+}$ and $p_{0,-}$ have degree two, $p_{2,+}$ and $p_{2,-}$ have degree three, and $p_{3,+}$ and $p_{3,-}$ have degrees eight and six, respectively. For example,

$$p_{0,+} = 9st^2 + (9 - 3s^2)t + 4s^3 - 12s, \quad p_{0,-} = (4s + 3t)(s^2 - 3st - 3).$$

The exact coefficient polynomials for all six factors are archived with the proof objects. The multiplicities in (9) are the symmetry multiplicities, not numerical eigenvalue clustering; the t^3 factor is the translation orbit. This factorization is a structural identity. It also gives an exact continuation proof of the signs. Indeed, after removal of the displayed t^3 , the constant term of every factor in (9) has no root on $0 < s < 1$ by exact Sturm counts in s^2 . Thus no nontranslation generalized eigenvalue crosses zero. At the rational sample $s = 1/2$, exact Sturm counts in t give

$$(n_+, n_-, n_0) = (1, 60, 3)$$

for the full generalized Hessian. The three zeros are precisely the translation orbit: its enstrophy-metric Gram determinant is

$$\det G_{\text{trans}} = \frac{8(3 + s^2)^3}{729} > 0.$$

Every nontrivial isotypic sector is $\mathcal{E}$ -tangent; in the remaining trivial-even sector, the $\mathcal{E}$ -tangent direction $w = (16b, -a)$ has exact curvature

$$w^T H_s w = \frac{32(s^2 - 3)(s^2 + 1)}{3s} < 0.$$

Since $D_E^{-1}H_s$ is similar to a real symmetric matrix, its spectrum is real and continuous in s . Consequently the constrained Hessian is strictly negative off translations for every $0 < s < 1$. The interval cover below is a separate Arb cross-check of this symbolic continuation argument.

For $0 \leq s \leq 47/50$, an exact shell-aware congruence removes the apparent $1/s$ singularity of the bordered Hessian: the $|k|^2 = 1$ control coordinates are scaled by s^{-1} , the bordered matrix is multiplied by s , and the resulting entries are simplified algebraically before interval evaluation. A cover by 188 rational intervals separately certifies inertia $(4+, 64-, 0)$ of this continuous rescaled form. The direct 238-interval cover of $9049/10000 \leq s \leq 1$ certifies the same inertia. The covers overlap, since $47/50 > 9049/10000$. Both covers were evaluated separately with 256- and 512-bit Arb arithmetic. For $s > 0$, the rescaled form is congruent to the ordinary bordered Hessian and has the same inertia.

Theorem 4.2 (Uniform competing branch). *For every $\mathcal{E}, \nu > 0$, the branch (7) is a strict local maximum of $\mathcal{R}$, modulo translations, in the complete 64-real-coordinate Fourier class. This is not a global-maximality claim.*

At $(\nu, \mathcal{E}) = (1/10, 100)$, the certified rate is 417.844001666639..., exceeding the six-amplitude rate 411.566601974923... at the same $\eta = 10^4$.

Theorem 4.3 (Competing full-space local branch). *At $(\nu, \mathcal{E}) = (1/10, 100)$, the branch (7) is a strict local maximum of $\mathcal{R}$ modulo translations in the complete 64-real-coordinate Fourier class. Its rate is strictly greater than that of the six-amplitude branch at the same parameter. It is not asserted to be a global maximum of the 64-coordinate problem.*

Theorem 4.4 (Exact symmetry-branch switch). *On the high- η branch in (4), the rates of the two symmetry branches cross exactly once, at*

$$\eta_{\times} = 27992.671985287174015090 \dots$$

The two-amplitude rate is larger for $1867/4 < \eta < \eta_{\times}$, and the six-amplitude rate is larger for $\eta > \eta_{\times}$.

Proposition 4.5 (Exact high-parameter efficiency). *Along the two symmetry branches, the normalized optimized rates have the limits*

$$\lim_{\eta \rightarrow \infty} \frac{\mathcal{R}_{\text{six}}}{\nu^3 \eta^{3/2}} = \frac{8\sqrt{138}}{207}, \quad \lim_{\eta \rightarrow \infty} \frac{\mathcal{R}_{\text{two}}}{\nu^3 \eta^{3/2}} = \frac{4}{9}.$$

The first is strictly larger: the difference of their squared values is $16/1863$. This comparison concerns only the two symmetry branches.

Proof. The rational six-amplitude parameterization has $\eta(\mu) \sim (69/32)\mu^2$ and $\mathcal{R}_{\text{six}}/\nu^3 \sim (23/16)\mu^3$, giving the first limit. The closed formula (8) gives the second directly. Squaring the two positive limits yields $8832/42849$ and $16/81$, whose difference is $16/1863 > 0$. $\square$

Proposition 4.6 (Exact inviscid six-amplitude extremizer). *At unit enstrophy, the high- η six-amplitude branch converges, after the fixed sign gauge, to*

$$(a, b, c, d, e, f) = \left(\frac{\sqrt{69}}{23}, -\frac{\sqrt{690}}{552}, \frac{\sqrt{345}}{276}, \frac{\sqrt{690}}{138}, \frac{\sqrt{69}}{276}, \frac{\sqrt{138}}{69} \right).$$

This field has $\mathcal{E} = 1$, $P = 148/69$, and

$$A = \frac{8\sqrt{138}}{207}.$$

It is a global maximizer of A under $\mathcal{E} = 1$ within the six-amplitude fixed space. No full 64-coordinate inviscid cubic-globality statement is made.

Proof. The exact rational branch parameterization gives $\eta \sim (69/32)\mu^2$, hence $\nu\mu \rightarrow \sqrt{32/69}$ at unit enstrophy. Substitution into the squared-amplitude formulas and the recovery equations for b, c, e gives the displayed signed vector; direct substitution in (1)–(3) gives its three invariants. For any unit-enstrophy y in this fixed space, Theorem 3.2 gives

$$A(y) - 2\nu P(y) \leq \mathcal{R}(x_{\nu}),$$

where x_{ν} is the high- η branch. Letting $\nu \rightarrow 0^+$ is legitimate in this finite-dimensional compact constraint set and yields $A(y) \leq 8\sqrt{138}/207$. $\square$

Theorem 4.7 (Full-space local maximality at the inviscid endpoint). *Let x_* be the unit-enstrophy six-amplitude field in Proposition 4.6. For every $E_0 > 0$, a translated copy of $\sqrt{E_0}x_*$ is a strict local maximizer of A subject to $\mathcal{E} = E_0$ in the full 64-coordinate class, modulo translations. This is a local statement; it does not assert the full-space global cubic inequality.*

Proof. At $E_0 = 100$, translate the displayed field by $(\pi/2, 0, 0)$ and use the three phase constraints associated with the waves $(1, -1, -1)$, $(1, -1, 1)$, and $(1, 1, 0)$. The exact radical formula lies in a unique Krawczyk box at both 256 and 512 bits. The bordered Hessian has certified inertia $(4^+, 64^-, 0)$, and the phase slice is regular. Thus it is a strict local constrained maximum modulo translations. The archived proof object is `inviscid_six_full64_local_certificate.json`. Positive scaling is a bijection between the fixed-enstrophy constraint manifolds and scales the cubic A by a positive factor, which gives every $E_0 > 0$. $\square$

Theorem 4.8 (Exact inviscid coexistence of two full-space local maxima). *For every $E_0 > 0$, the full 64-coordinate inviscid cubic problem has at least two distinct strict local maxima modulo translations. They are the six-amplitude branch in Theorem 4.7, with*

$$A_{\text{six}} = \frac{8\sqrt{138}}{207}E_0^{3/2},$$

and the two-amplitude competing branch, with

$$A_{\text{two}} = \frac{4}{9}E_0^{3/2}.$$

They have the strict exact separation

$$A_{\text{six}} - A_{\text{two}} = \frac{8\sqrt{138} - 92}{207}E_0^{3/2} > 0.$$

No assertion is made that these exhaust all full-space local maxima.

Proof. The first local maximum is Theorem 4.7. For the second, the inviscid formula (7) gives $(a, b) = (\sqrt{2E_0}/3, \sqrt{E_0}/12)$ and direct substitution in (5) gives A_{two} . At $E_0 = 100$, after translation by $(0, \pi/2, 0)$, the exact formula lies in a unique Krawczyk box at both 256 and 512 bits, with regular phase slice and bordered inertia $(4^+, 64^-, 0)$; the corresponding proof artifact is listed below. Positive scaling gives all $E_0 > 0$. Finally, the displayed difference is positive because $8\sqrt{138} > 92$. $\square$

The efficiency gap becomes visible only at a surprisingly large parameter because the next asymptotic term favors the two-amplitude branch. More precisely, with $t = \eta^{-1/2}$, exact inversion of the six-amplitude rational parameterization gives the following expansion. To justify the inversion, put $z = t\mu$ and clear denominators in $t^2\eta(\mu) = 1$. At $t = 0$ the resulting equation is $69z^2/32 = 1$, whose positive root $z_0 = 4\sqrt{138}/69$ is simple. The analytic implicit-function theorem therefore applies at $(z_0, 0)$; coefficient comparison gives

$$z = \frac{4\sqrt{138}}{69} - \frac{296}{69}t + \frac{1913\sqrt{138}}{9522}t^2 + O(t^3).$$

Substitution into the exact rational rate gives

$$\frac{\mathcal{R}_{\text{six}}}{\nu^3\eta^{3/2}} = \frac{8\sqrt{138}}{207} - \frac{296}{69}t + \frac{1913\sqrt{138}}{4761}t^2 + O(t^3),$$

whereas (8) gives

$$\frac{\mathcal{R}_{\text{two}}}{\nu^3\eta^{3/2}} = \frac{4}{9} - \frac{8}{3}t + \frac{2}{3}t^2 + O(t^3).$$

Thus the first two terms predict a crossover at

$$\eta_{\text{two-term}} = \frac{7056}{(23 - 2\sqrt{138})^2} = 28834.3012\dots,$$

close to, but not used to prove, the interval-certified value in Theorem 4.4.

Proof certificate. Eliminating b from equality of rate and η gives a degree-12 resultant whose leading coefficient is positive and whose other coefficients are negative. It has exactly one positive root by Descartes' rule. The map $\eta(\mu)$ on the six-amplitude branch has positive derivative. A two-variable Krawczyk calculation encloses the crossing at both 256 and 512 bits; interval rate comparisons at $\eta = 10^4$ and 4×10^4 establish the two signs. The machine-checkable polynomial and enclosures are archived with the reproducibility artifacts listed below. $\square$

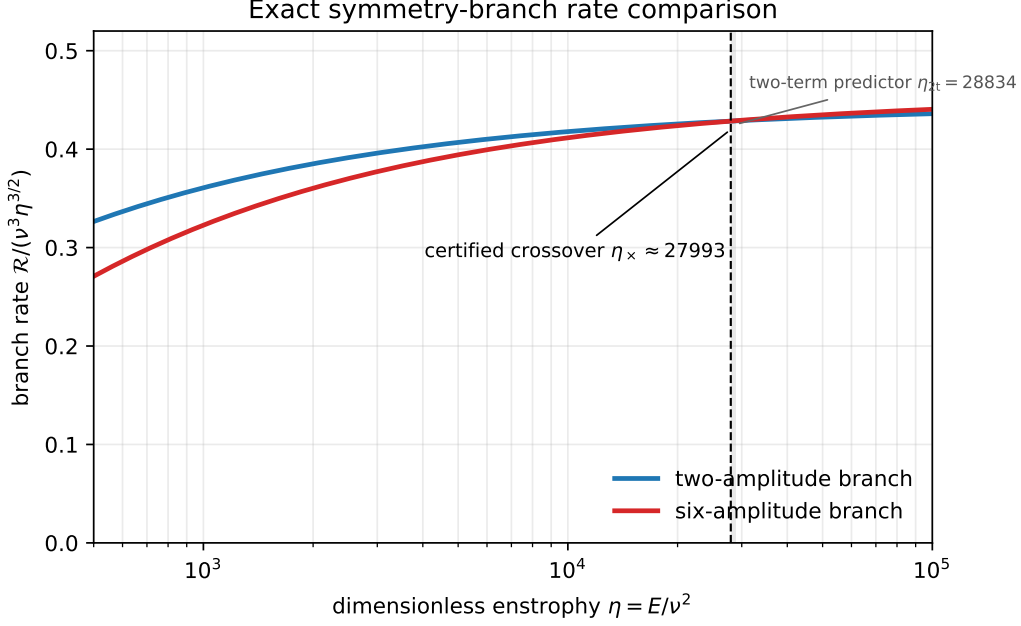


Figure 1: Exact reduced rate formulas for the two symmetry branches, normalized by $\nu^3 \eta^{3/2}$. The vertical line marks the interval-certified crossover from Theorem 4.4; the adjacent gray sliver and annotation mark the two-term asymptotic predictor, which is not used in that certification. The plot compares only the two reduced branches; it is not an ordering of all 64-coordinate fields.

5 Conclusion

The fixed-entrophy instantaneous-production problem contains more exact structure than is visible from numerical multistart optimization alone. In a specified 16-pair Fourier class, we identify an ansatz-global six-amplitude branch and a distinct exact two-amplitude full-KKT branch. At high parameter, the latter is rigorously a higher full-space local maximum at a displayed point, while the inviscid endpoint of the former is also a rigorously certified full-space local maximum; moreover its entire high- η branch is uniformly locally maximal in the full Fourier class. At the inviscid endpoint the two branches coexist as distinct full-space local maxima with an exact positive value gap. Proposition 2.1 adds an exact rational bound for one shellwise component of stretching, rather than a global bound on the full cubic. The two viscous branch rates undergo one certified switch. The result is a finite-dimensional computer-assisted variational theorem. Its main value is to expose and certify distinct mechanisms in a standard extreme-flow problem, not to infer anything about arbitrary Fourier data or long-time PDE dynamics.

6 Reproducibility and scope

The repository contains exact rational Fourier algebra, source hashes, dependency manifests, and separate 256/512-bit Arb records. Distinct precisions protect against precision-specific numerical accidents, but they are not a substitute for an independent implementation audit. The command

```
python scripts/verify_paper_a.py
```

recomputes the theorem claims, including the two-amplitude full-KKT branch and the unique crossover. The finite five-point sweep is retained as a cross-check. The separate uniform certificate

combines 188 rescaled low- s intervals with 238 direct high- s intervals and proves full-space local maximality of the competing branch for every positive η ; neither certificate makes a full-space global claim. An additional 2,048-box compact interval-continuation certificate proves the same full-space local statement for the high- η six-amplitude branch; again it is a local finite-dimensional theorem only. Proposition 2.1 is independently exact: its rational Gram identity verifies one shellwise cubic bound, but does not supply a global bound on the full stretching term.

None of the results proves a full-space global maximum, extends beyond this Fourier cutoff, or addresses a finite-time singularity or global regularity.

Artifact list. The content-addressed release manifest in the `artifacts/proofs/` directory binds every checker, candidate, source, manuscript file, and proof object by SHA-256. The primary proof objects are:

```
reduced_static_global_arb_certificate.json
reduced_static_standalone_algebra_audit.json
high_branch_eta10000_arb_certificate.json
inviscid_six_full64_local_certificate.json
inviscid_competing_full64_local_certificate.json
six_branch_high_eta_uniform.json
shell112_sos_certificate.json
shell123_sos_certificate.json
shell224_sos_certificate.json
shell334_sos_certificate.json
reduced_competing_formula.json
competing_symmetry_group.json
competing_representation_sectors.json
competing_high_eta_uniform.json
competing_all_eta_uniform.json
symmetry_branch_crossover.json
```

The one-command verifier is `scripts/verify_paper_a.py`. The reviewer guide in the `paper/` directory gives the claim-to-artifact map, including the exact symbolic spectral-continuation record and the separate Arb interval records.

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